PLK2 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes PLK2 gene fusions from the msk_impact_50k_2026 study, covering 7 fusion events. 2/4 fusions with known frame status (50.0%) were in-frame. 4/7 fusions (57.1%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=0.285714, not statistically significant at alpha=0.05). No literature benchmark is configured for PLK2.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 7 analyzed fusion events, identifying 4 distinct fusion partner genes. The most frequent partner was PDE4D, observed in 4 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 2/2 (100.0%) of in-frame fusions retained the domain, p=0.285714 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.237624.

Domain disruption

Domain-disruption analysis was skipped: PLK2 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

100% of 1 fusions with this gene as the 3' partner join at exon 1, against 9% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.851485, corrected for testing every exon, not statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 7 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 238 aa (exon 5/6 boundary; label-permutation p=0.414585). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -96 aa from the nearest configured domain boundary at 334 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=4) and "not_retained" (n=3). For Frame_status, retained: 2/3 (66.7%, 95% CI 20.8%-93.9%); not_retained: 0/1 (0.0%,

95% CI 0.0%-79.3%). Welch's t-test comparing Tumor_variant_count between groups gave means of 24.75 vs 34.3333, p=0.754733 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for PLK2 in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (5 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 0 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: PLK2 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: PLK2 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection reported: no candidate window of the configured widths (25, 50, 100, 200) met the minimum-event-count guard (min_events_per_window=4) on both sides of the split

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PDE4D	4	0.5714285714285714	0.06241101320710366	None	None	0.23785042994984074	confidence_certainty, domain_retention, recurrence	0.32433287342358674	1
MYO15A	1	0.14285714285714285	0.06241101320710366	None	None	0.23785042994984074	confidence_certainty, domain_retention, recurrence	0.12653067562138898	2
PPP2R3B	1	0.14285714285714285	0.06241101320710366	None	None	0.23785042994984074	confidence_certainty, domain_retention, recurrence	0.12653067562138898	3
WWOX	1	0.14285714285714285	0.06241101320710366	None	None	0.23785042994984074	confidence_certainty, domain_retention, recurrence	0.12653067562138898	4

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
1	1	3	0	3	None	1.0	-0.0
126	1	3	1	2	0.6666666666666666	1.0	-0.0
196	1	3	2	1	0.16666666666666666	0.48571428571428565	0.3136191229720018
238	1	3	3	0	0.0	0.14285714285714285	0.8450980400142568
336	2	2	3	0	0.0	0.42857142857142855	0.36797678529459443
386	3	1	3	0	0.0	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

2	2
0	3

frame_domain_contingency_tables_by_tumor_type

stratum	table
BRCA	[0, 0], [0, 1]
BRCNOS	[1, 0], [0, 0]
GB	[0, 1], [0, 0]
GEJ	[0, 1], [0, 0]
LMS	[0, 0], [0, 1]
PRAD	[0, 0], [0, 1]
ULMS	[1, 0], [0, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	7	4	3	0	0.41534391534391535

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
1	7	7	1.0

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
MYO15A	1	1	1	1
PDE4D	4	4	4	4
PPP2R3B	1	1	1	1
WWOX	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
5	57752000	57753000	2	EVT-P-0027130-T01-IM6-8, EVT-P-0032779-T01-IM6-3

chromosome	bin_start	bin_end	n_events	event_ids
5	57754000	57755000	2	EVT-P-0005216-T01-IM5-6, EVT-P-0060865-T01-IM7-9
5	57751000	57752000	1	EVT-P-0005582-T01-IM5-7
5	57753000	57754000	1	EVT-P-0005900-T02-IM6-22
5	57755000	57756000	1	EVT-P-0033104-T01-IM6-20

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	2	1	6	0.16666666666666666	0.0399745182353878	0.6633050918890576
five_prime	4	1	6	0.16666666666666666	0.056856187290969896	0.5284583304185184
five_prime	5	1	6	0.16666666666666666	0.09364548494983277	0.35101176572570303
five_prime	7	1	6	0.16666666666666666	0.045707915273132664	0.6112686465971436
five_prime	8	1	6	0.16666666666666666	0.08010829749960184	0.40442317229907326
five_prime	11	1	6	0.16666666666666666	0.05813027552157987	0.520200216935409
three_prime	1	1	1	1.0	0.09093804745978659	1.041254374812501

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032779-T01-IM6-3	P-0032779-T01-IM6	P-0032779	Breast Cancer	BRCNOS	MYO15A-PLK2	MYO15A	in-frame	five_prime	57752959	7	336	True	336	True	exact match	retained	1.0	False	Protein kinase domain			PLK2 (NM_006622) -MYO15A (NM_016239) Rearrangement : t(5;17)(q11.2;p11.2)(chr5:g.57752959::chr17:g.18037660) | Protein Fusion: in frame {PLK2:MYO15A}	NA
EVT-P-0005216-T01-IM5-6	P-0005216-T01-IM5	P-0005216	Prostate Cancer	PRAD	PLK2-PDE4D	PDE4D	out-of-frame	five_prime	57754794	2	126	True	126	True	exact match	disrupted	0.174603174603 1746	True			Protein kinase domain	PLK2 (NM_006622) -PDE4D (NM_001104631) Rearrangement: c.378+18_c.455+337033dup | Protein fusion: out of frame (PLK2-PDE4D)	NA

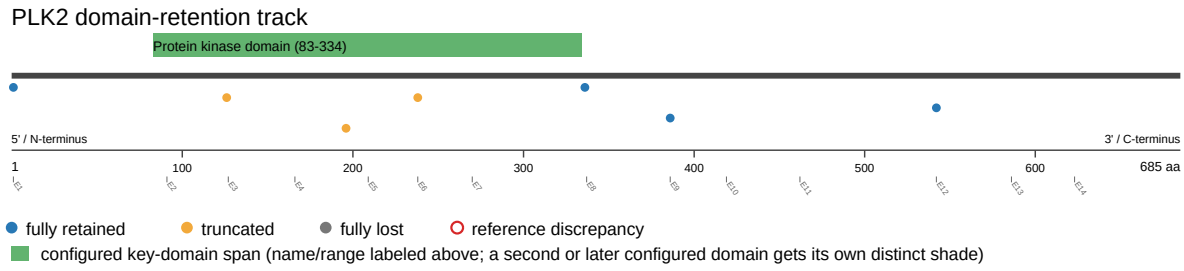
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005582-T01-IM5-7	P-0005582-T01-IM5	P-0005582	Uterine Sarcoma	ULMS	PLK2-PDE4D	PDE4D	in-frame	five_prime	57751339	11	542	True	541	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain			PLK2 (NM_006622) -PDE4D (NM_001104631) rearrangement: c.1625+27: PLK2_c.456-334412: PDE4Ddup | Protein fusion: in frame (PLK2-PDE4D)	NA
EVT-P-0027130-T01-IM6-8	P-0027130-T01-IM6	P-0027130	Esophagogastric Cancer	GEJ	PLK2-PDE4D	PDE4D	out-of-frame	five_prime	57752758	8	386	True	385	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain			PLK2 (NM_006622) - PDE4D (NM_001104631) rearrangement: c.1156+14:PLK2_c.922-15282:PDE4Ddup | Protein Fusion: out of frame {PLK2:PDE4D}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0060865-T01-IM7-9	P-0060865-T01-IM7	P-0060865	Soft Tissue Sarcoma	LMS	PLK2-PDE4D	PDE4D	unknown	five_prime	57754264	4	196	False	195	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.452380952380 9524	True			Protein kinase domain	PLK2 (NM_006622) - PDE4D (NM_001104631) rearrangement: c.587:PLK2_c.808+47369:PDE4D dup | Protein Fusion: mid-exon {PLK2:PDE4D}	NA
EVT-P-0033104-T01-IM6-20	P-0033104-T01-IM6	P-0033104	Glioma	GB	PPP2R3B-PLK2	PPP2R3B	unknown	three_prime	57755878	1	1	True	1	True	exact match	retained	1.0	False	Protein kinase domain			PPP2R3B (NM_013239) - PLK2 (NM_006622) Rearrangement : t(X,5)(p22.32;q11.2)(chrX:g.309911::chr5:g.57755878) | Transcript Fusion { PPP2R3B:PLK2}	NA

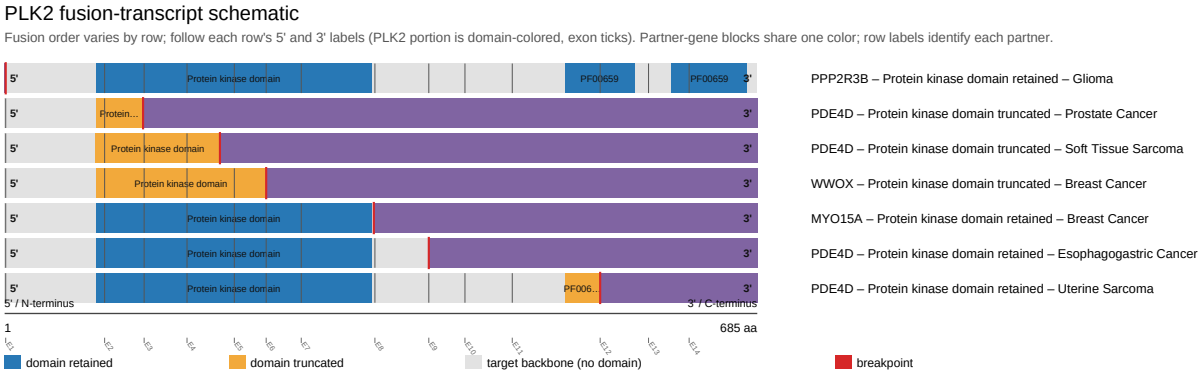
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005900-T02-IM6-22	P-0005900-T02-IM6	P-0005900	Breast Cancer	BRCA	WWOX-PLK2	WWOX	unknown	five_prime	57753806	5	238	True	237	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.619047619047619 1	True			Protein kinase domain	WWOX (NM_016373) - PLK2 (NM_006622) rearrangement: t(5,16)(q11.2,q23.1)(chr5:g.57753806::chr16:g.78828029) | Antisense Fusion	NA

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.